



```
31 # -----
32
33 scaler = StandardScaler()
34
35 scaled_data = scaler.fit_transform(df)
36
37 print("\nDATA STANDARDIZED")
38
39 # -----
40 # PCA
41 # -----
42
43 pca = PCA(n_components=2)
44
45 pca_data = pca.fit_transform(scaled_data)
46
47 print("\nPCA COMPLETED")
48 print("Original Shape:", scaled_data.shape)
49 print("Reduced Shape:", pca_data.shape)
50
51 # -----
52 # ELBOW METHOD
53 # -----
54
55 print("\nELBOW METHOD")
56
57 wcss = []
58
59 for i in range(1, 11):
60
```

```
object3.py 7
51 # -----
52 # ELBOW METHOD
53 # -----
54
55 print("\nELBOW METHOD")
56
57 wcss = []
58
59 for i in range(1, 11):
60     kmeans = KMeans(
61         n_clusters=i,
62         random_state=42
63     )
64     kmeans.fit(pca_data)
65     wcss.append(kmeans.inertia_)
66     print("Clusters:", i,
67         | "WCSS:", round(kmeans.inertia_, 2))
68
69 # -----
70 # SILHOUETTE SCORE
71 # -----
72
73 print("\nSILHOUETTE SCORES")
74
75 for i in range(2, 11):
76     kmeans = KMeans(
77         n_clusters=i
```

```
72 # -----
73 # SILHOUETTE SCORE
74 # -----
75
76 print("\nSILHOUETTE SCORES")
77
78 for i in range(2, 11):
79     kmeans = KMeans(
80         n_clusters=i,
81         random_state=42
82     )
83     labels = kmeans.fit_predict(pca_data)
84     score = silhouette_score(
85         pca_data,
86         labels
87     )
88     print("Clusters:",
89           i,
90           "Score:",
91           round(score, 4))
92
93 # -----
94 # FINAL K-MEANS
95 # -----
96
97 kmeans = KMeans(
98     n_clusters=4,
```

```
ER_SEGMENTATI...
3.py 7
98 # -----
99 # FINAL K-MEANS
100 # -----
101
102 kmeans = KMeans(
103     n_clusters=4,
104     random_state=42
105 )
106
107 clusters = kmeans.fit_predict(
108     pca_data
109 )
110
111 df['cluster'] = clusters
112
113 print("\nCLUSTER COUNTS")
114 print(df['cluster'].value_counts())
115
116 # -----
117 # CUSTOMER PERSONAS
118 # -----
119
120 print("\nCUSTOMER PERSONAS")
121
122 print("Cluster 0 : Budget Customers")
123 print("Cluster 1 : Premium Customers")
124 print("Cluster 2 : Regular Customers")
125 print("Cluster 3 : High Value Customers")
126
127 print("\nPROJECT COMPLETED SUCCESSFULLY")
```



```
customer_segmentation...
ect3.py 7

project3.py > ...
PROBLEMS 7 OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, lexcludeText, t... Code
[Running] python -u "c:\Users\Keerthi Reddy\Desktop\Customer_Segmentation_Project\project3.py"
FIRST 5 RECORDS
|  |  | 0      1      2      ...      17      18      19
0  2.280887 -6.254320 -4.641365 ... -7.598072 3.526123 -2.114969
1  -0.849932 -4.371952 5.609354 ... -7.720602 -1.915156 -6.418569
2  -2.009513 9.486288 4.715834 ... 1.216691 -1.591966 -2.722157
3  -1.801535 -2.896369 7.581001 ... -8.608833 -3.233354 -8.189647
4  -1.707018 -5.733040 8.231073 ... -9.300667 -3.321948 -7.383098

[5 rows x 20 columns]

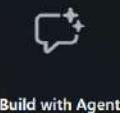
DATASET SHAPE
(1000, 20)

DATA STANDARDIZED

PCA COMPLETED
Original Shape: (1000, 20)
Reduced Shape: (1000, 2)

ELBOW METHOD
Clusters: 1 WCSS: 15513.04
Clusters: 2 WCSS: 7413.06
Clusters: 3 WCSS: 2072.65
Clusters: 4 WCSS: 111.77
Clusters: 5 WCSS: 100.14
Clusters: 6 WCSS: 88.85
Clusters: 7 WCSS: 77.85
Clusters: 8 WCSS: 68.05
Clusters: 9 WCSS: 62.81
Clusters: 10 WCSS: 57.61

SILHOUETTE SCORES
Clusters: 2 Score: 0.5989
```



Build with Agent

AI responses may be inaccurate.

[Generate Agent Instructions](#) to onboard AI onto your codebase.

project3.py

Describe what to build

+ | | Auto |

Default A

CUSTOMER_SEGMENTATI...
project3.py 7

project3.py > ...

PROBLEMS 7

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Filter (e.g. text, excludeText, t...

Code

```
Clusters: 0 WCSS: 68.63
Clusters: 7 WCSS: 77.85
Clusters: 8 WCSS: 68.05
Clusters: 9 WCSS: 62.81
Clusters: 10 WCSS: 57.61
```

SILHOUETTE SCORES

```
Clusters: 2 Score: 0.5989
Clusters: 3 Score: 0.7667
Clusters: 4 Score: 0.9086
Clusters: 5 Score: 0.7644
Clusters: 6 Score: 0.6302
Clusters: 7 Score: 0.4894
Clusters: 8 Score: 0.35
Clusters: 9 Score: 0.348
Clusters: 10 Score: 0.3498
```

CLUSTER COUNTS

```
3 250
2 250
1 250
0 250
```

Name: Cluster, dtype: int64

CUSTOMER PERSONAS

```
Cluster 0 : Budget Customers
Cluster 1 : Premium Customers
Cluster 2 : Regular Customers
Cluster 3 : High Value Customers
```

PROJECT COMPLETED SUCCESSFULLY

[Done] exited with code=0 in 39.561 seconds

OUTLINE

TIMELINE